

From Text to Behavior: Making Probabilistic AI Visible with Embodied Robots

A Classroom Experiment with Programmable Mini AI Robots



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Motivation

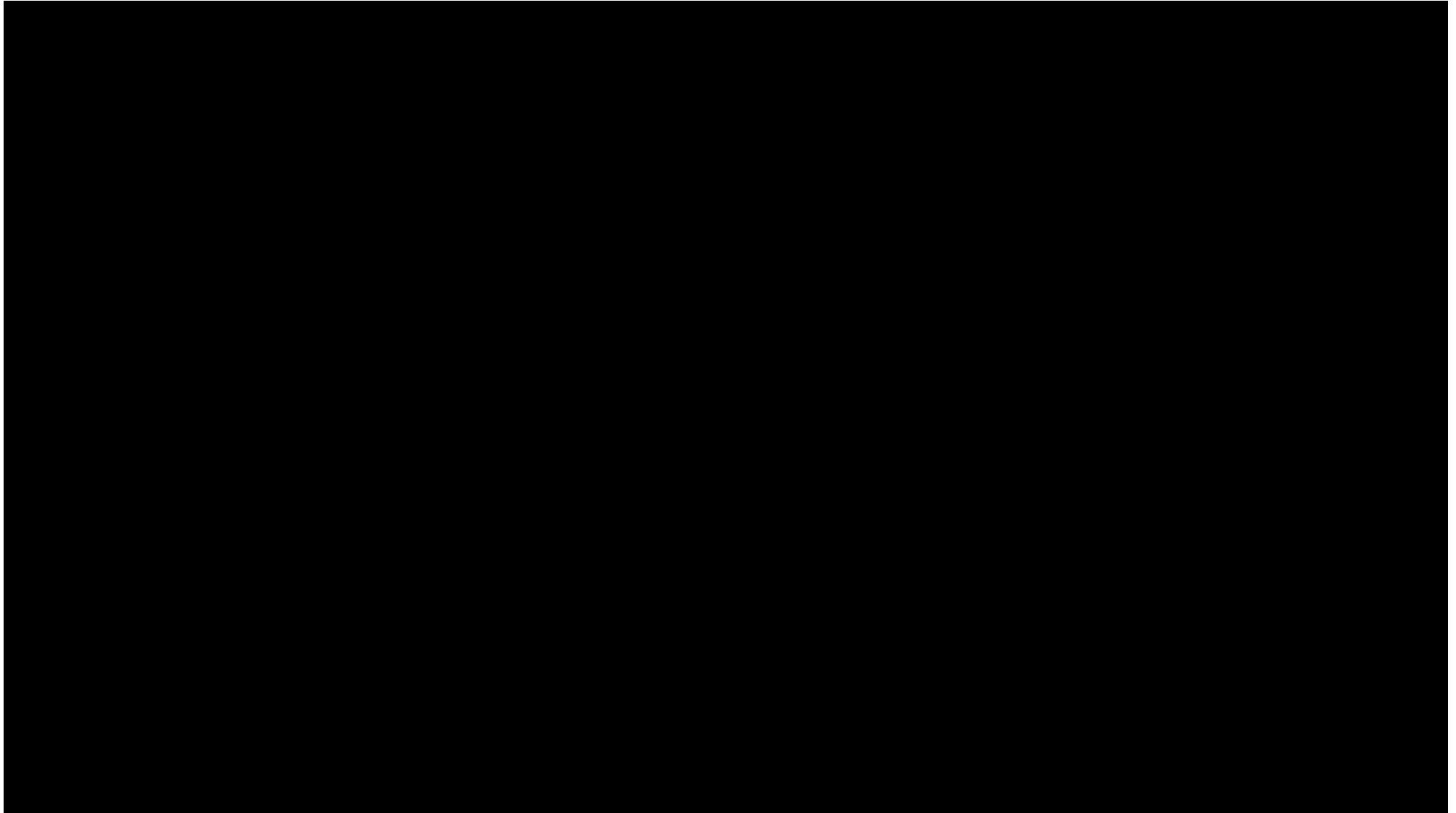
* Key Question

How can students reason about AI when **all they see is text on a screen?**

★ Answer: Give AI a Body

When AI **moves**, students can see probabilistic behavior instead of merely reading about it. Embodied feedback transforms an abstract concept into an observable, repeatable experiment.

The Frontier: Why Embodied AI Now?

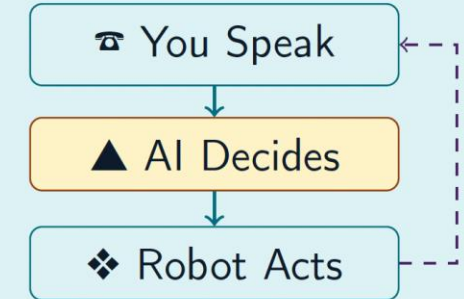


Project Overview

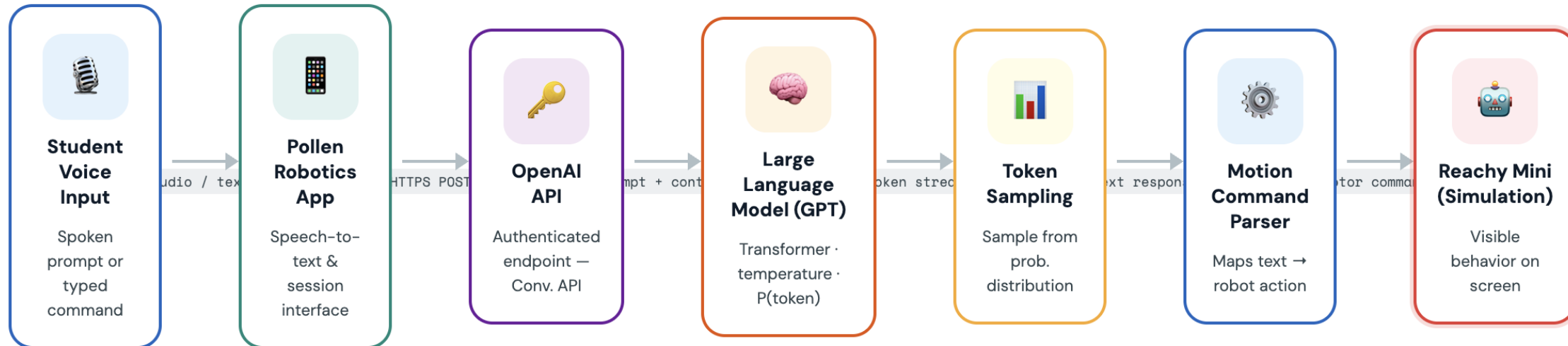
System Stack

- LLM APIs (OpenAI / Gemini)
- Rechy Mini robot simulation now • physical on order
- Real-time voice, vision & motion pipeline

* Learning Loop



The full decision loop becomes visible.



Learning Objectives

● What Students Should Be Able to Do

- Map prompts to robot behaviors
(input $\rightarrow$ output reasoning)
- Predict how modifying prompts changes robot actions
- Design prompts to produce a desired behavior
- Analyze failure cases and revise prompts



Assessment Methods

✓ How We Evaluate Learning

- Prompt design tasks
Students create prompts to achieve target behaviors
- Behavioral prediction questions
Given a prompt, predict robot action
- Failure analysis explanations
Explain why the system failed and how to fix it

★ Why It Matters

Students who can **analyze AI failure** are better equipped to **design reliable AI systems**.



Meet Reachy Mini



```
from reachy_mini import ReachyMini
from reachy_mini.utils import create_head_pose

with ReachyMini() as mini:
    # Look up and tilt head
    mini.goto_target(
        head=create_head_pose(z=10, roll=15, degrees=True, mm=True),
        duration=1.0
    )
```

What Students Learn by Watching the Robot

→ 1. Variability

Same question, different answers. Students observe **stochastic behavior** in real time.

✖ 2. Sensitivity

Small prompt changes cause **large behavioral shifts**, revealing how inputs shape outcomes.

👉 3. Failure

Students **intentionally confuse** the system and analyze what goes wrong.



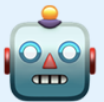
Variability: Same Prompt → Different Robot Behavior

EXAMPLE 01 • IDENTICAL PROMPT, THREE TRIALS

● STUDENT SAID

"Can you dance?"

Spoken 3 times — word for word identical



3 Different Motions

Trial 1 → Head moved back and forth

Trial 2 → Head rotated in a circle

Trial 3 → Head bobble

EXAMPLE 02 • TWO WORDS ADDED — MOTION BREAKS

● STUDENT SAID

"Can you spin"

then immediately

"Can you spin your head"

Two prompts, seconds apart, one extra phrase



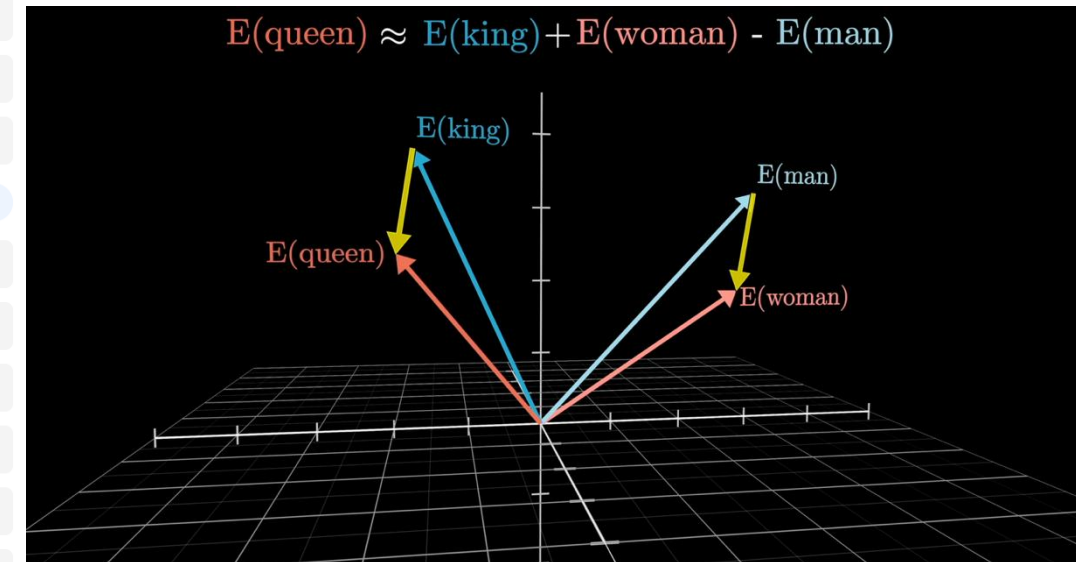
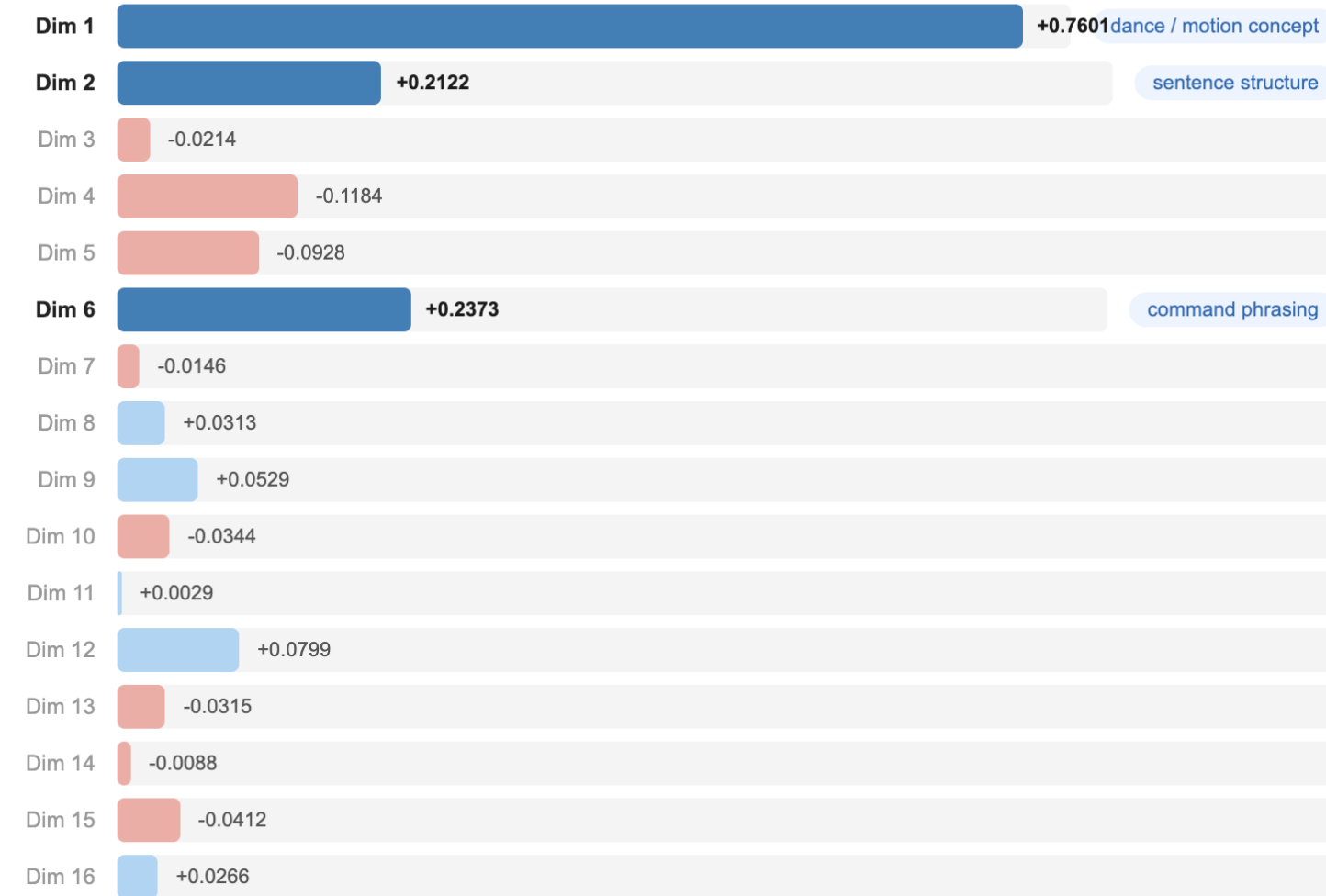
"Your head" broke it

Adding just two words shifted the token distribution enough that the robot produced no motion — despite identical intent.



Why Small Prompt Changes Lead to Different Robot Behaviors

"can you dance"



Sensitivity: Different Words Produce Completely Different Robot Behavior

EXAMPLE 01 • NEAR-SYNONYMS, OPPOSITE OUTCOMES

STUDENT A SAID "Can you hop?"	STUDENT B SAID "Can you jump?"
 Head rotated Robot mapped "hop" to a partial head movement	 No movement at all Robot did absolutely nothing

EXAMPLE 02 • CULTURAL REFERENCE SHAPES THE GESTURE

STUDENT A SAID "Dance like a rock star"	STUDENT B SAID "Dance like Michael Jackson"
 Nods its head Generic rock energy → simple head nod	 Side-to-side sway Specific icon → distinct signature move

Failure: Adversarial Prompts

EXAMPLE 01 • CONTRADICTION COMMAND

● STUDENT SAID

"Stand still, then dance."

Two physically contradictory instructions in one prompt



Ignored "stand still" — just danced

The robot started dancing immediately. The contradiction was never resolved — the model picked one instruction and silently dropped the other.

⚠️ CONTRADICTION FAILURE

EXAMPLE 02 • OUT-OF-SCOPE COMMAND

● STUDENT SAID

"Explain quantum physics."

A cognitive task — not a motion command



Refusal — No Action

The system could not translate an abstract knowledge request into physical motion. The robot stayed still with an unsure verbal response and zero movement.

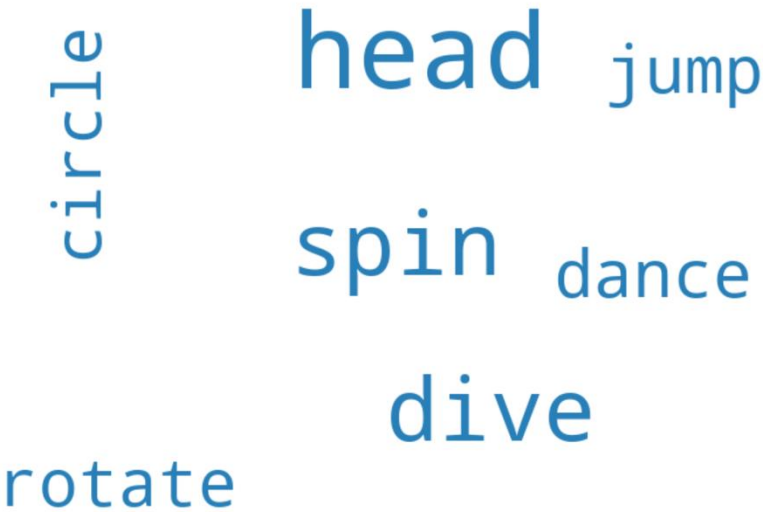
🚫 REFUSAL FAILURE



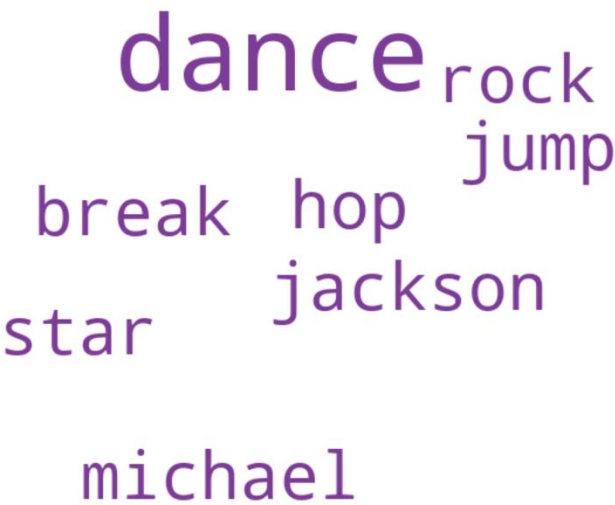
Student Prompt Strategies Across Learning Phases

Student Prompts — Word Clouds by Round

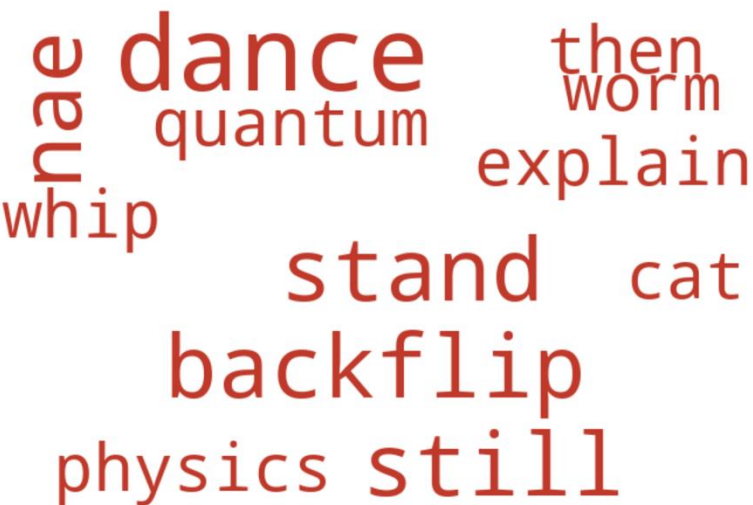
Variability (Round 1)



Sensitivity (Round 2)

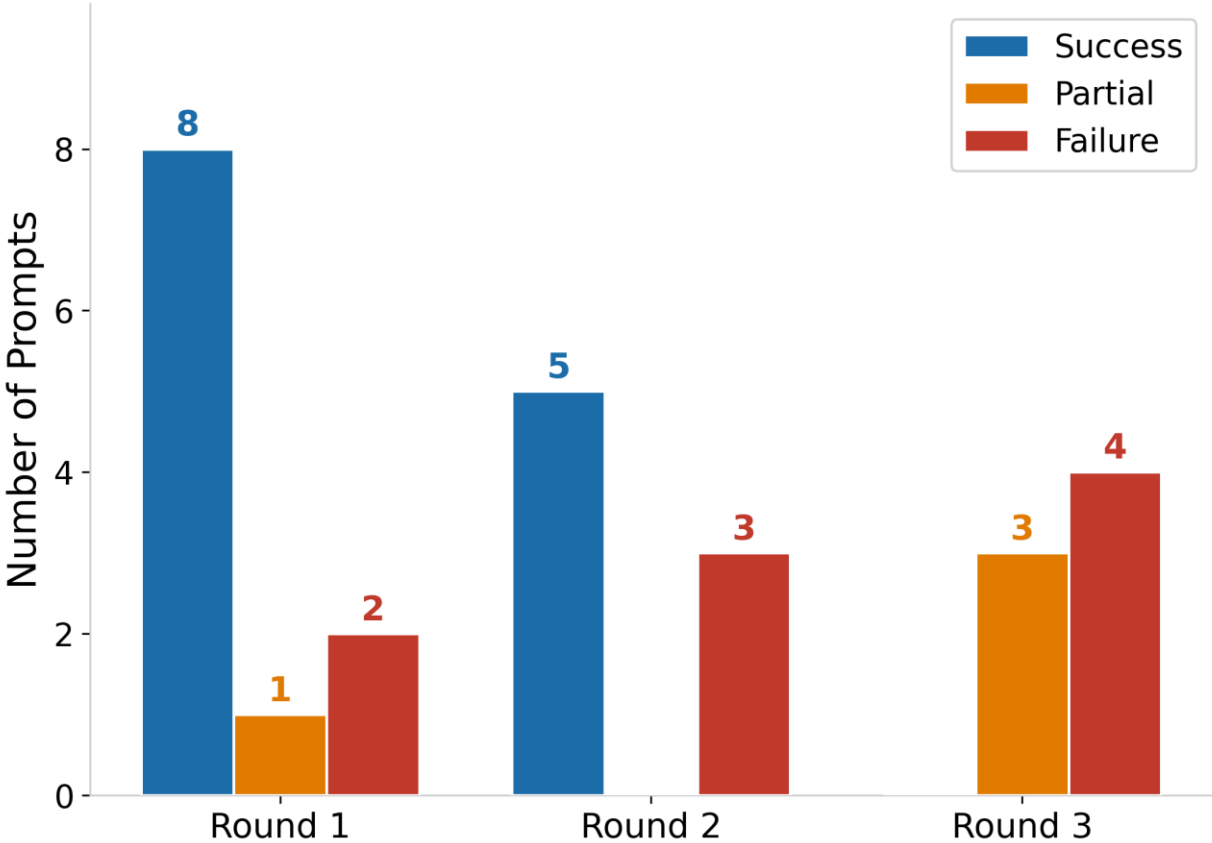


Failure Analysis (Round 3)

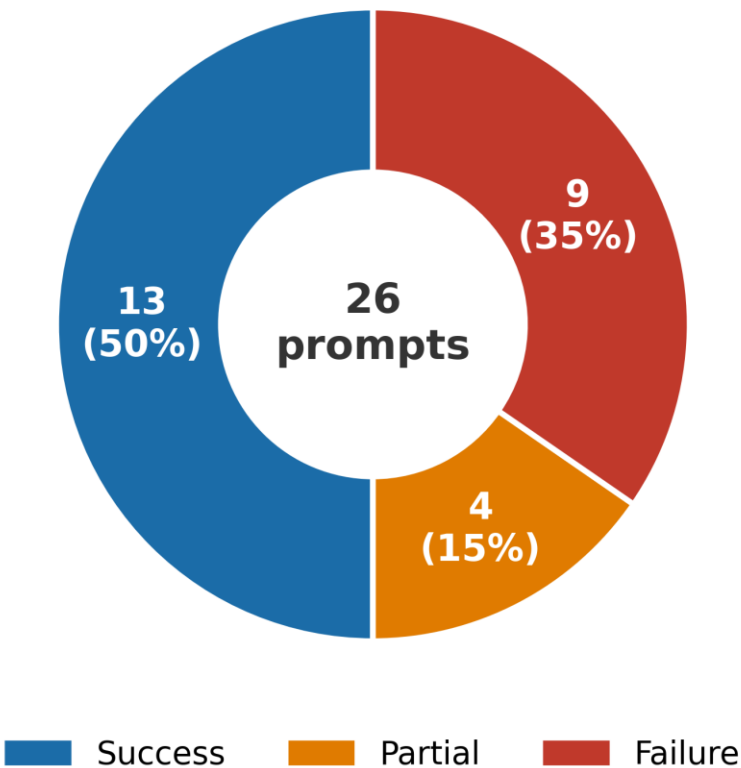


Learning Evidence: Student Control of Robot Behavior

Outcomes per Round

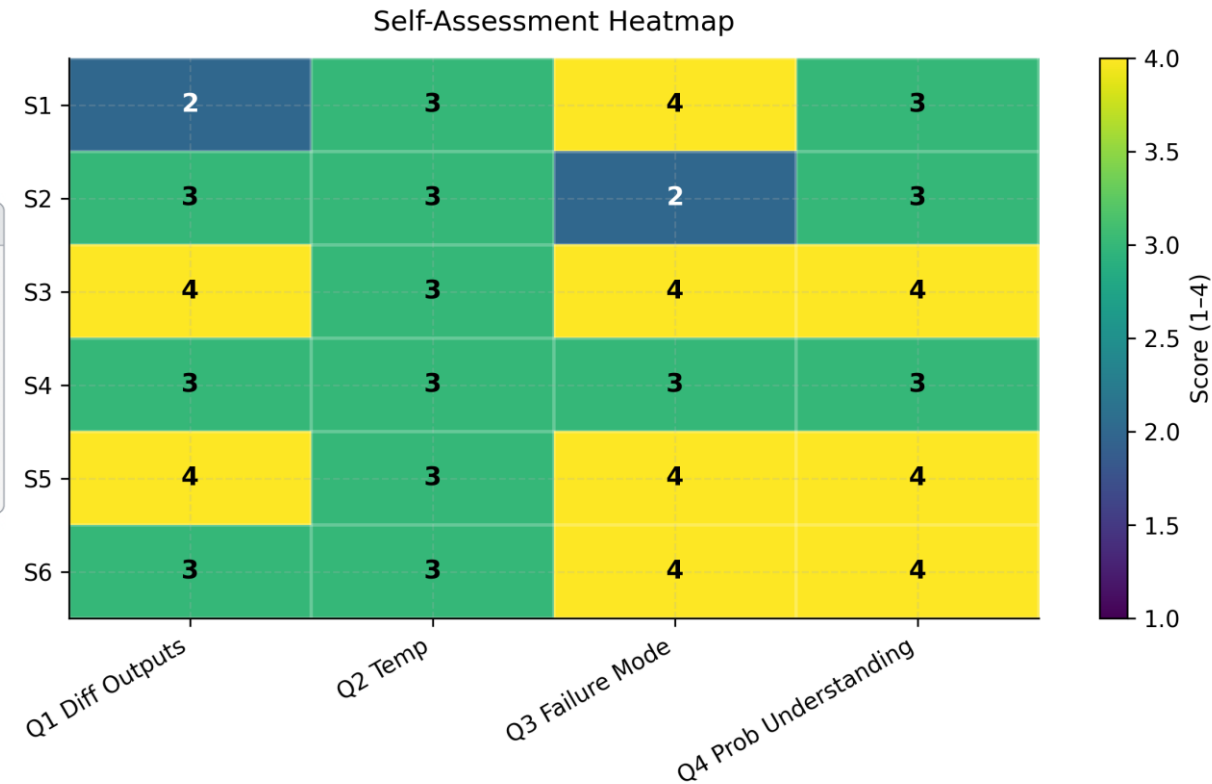


Overall Outcome Split



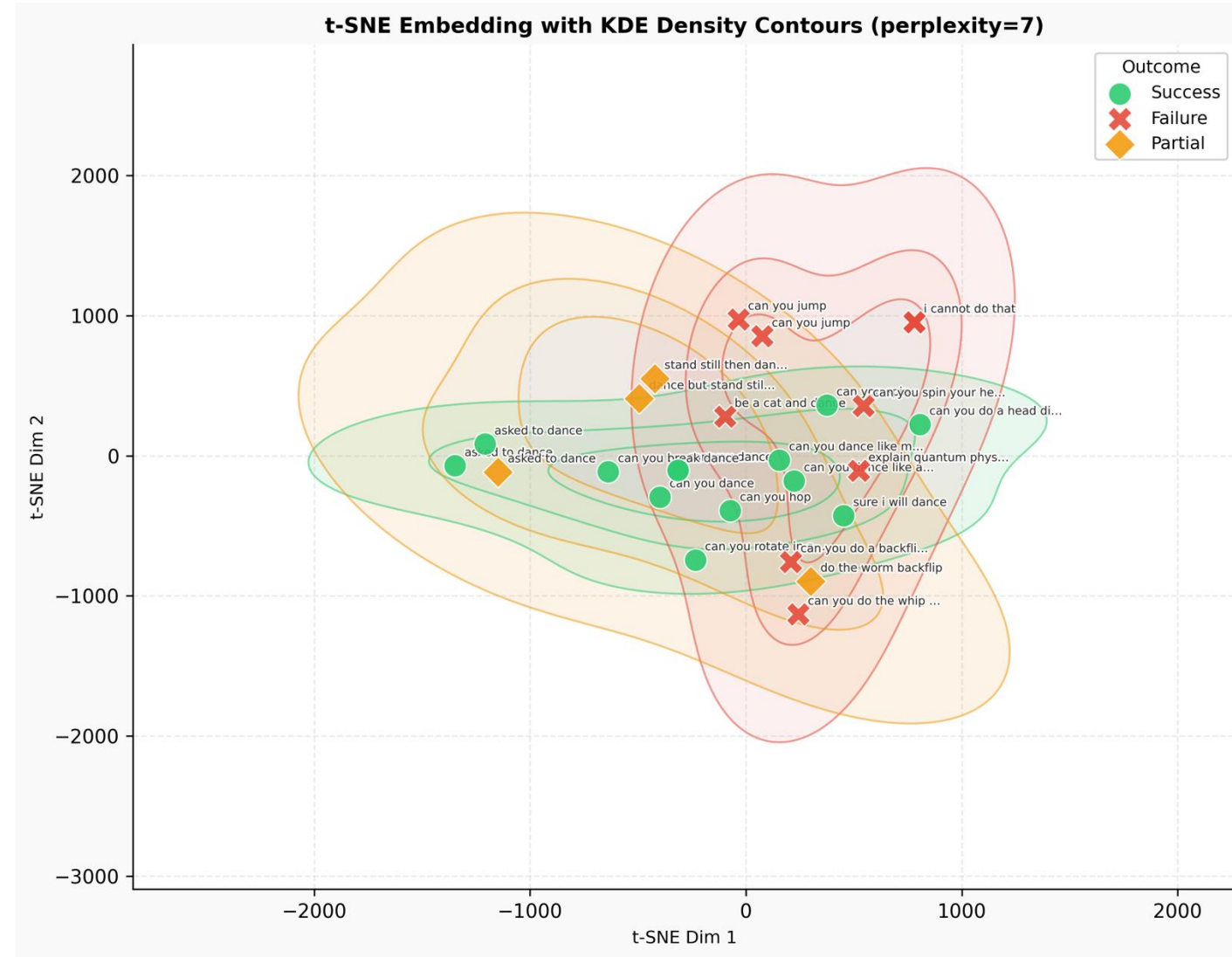
Student Learning Evidence (Self-Assessment)

Self-Assessment (circle one per row)				
	1 — Not at all	2	3	4 — Fully
I can explain why the same prompt gives different outputs	①	②	③	④
I understand what “temperature” does to model output	①	②	③	④
I can identify at least one AI failure mode	①	②	③	④
Seeing the robot helped me understand probability	①	②	③	④

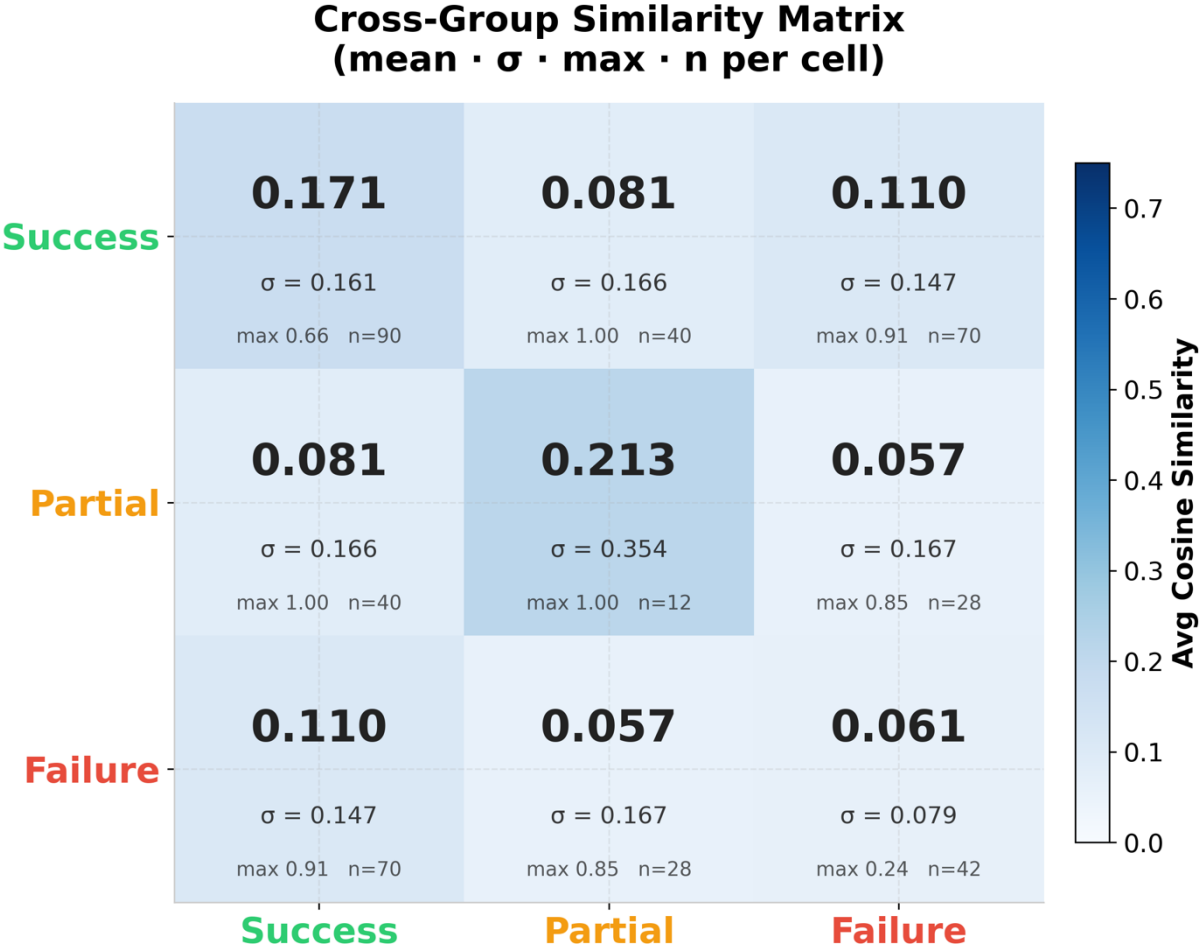


Why Robot Behavior Is Hard to Predict

- ❑ No clear boundaries between success, failure, and partial outcomes
- ❑ Success prompts were the most linguistically diverse



Quantifying Overlap in Behavior Space



All mean scores are below 0.22, confirming no group is semantically well-separated from the others

What Students Actually Learned

▲ Observed Learning Outcomes

- Predict robot behavior from prompt structure
- Identify ambiguous or conflicting prompts
- Intentionally manipulate prompts to test system limits
- Recognize that LLM outputs are **probabilistic**

★ Key Shift in Thinking

Students moved from “*asking questions*” → “**experimenting with an AI system**”



Impact

✧ For Students

From **passive observers** of AI outputs to **active analysts** of AI behavior.

✧ For the Curriculum

Embodied AI brings **human–AI interaction** into the data science curriculum, closing the gap between theory and lived experience.

LIVE NOW

Live Demonstration

(Simulation)

Ask the robot a question

Observe **variation**, **reasoning**, and **motion**

Watch probabilistic AI in action

